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# Landscape Connectivity along Balinsky's (1962) Drought corridor between eastern and southern Africa (Kalahari – Rift Valley Trans-Frontier Conservation Landscapes - KALARIVA - TFCL): An option for climate change adaptation by African wildlife

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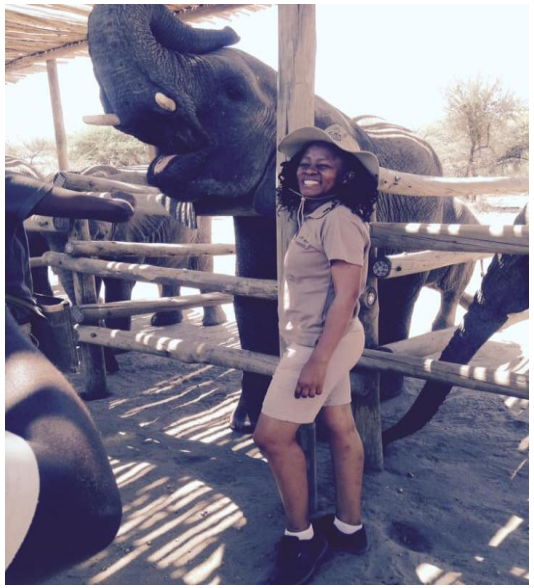


By

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Department of Environmental Science, University of Botswana


10 March 2026; Skukuza, Krunger National Park, South Africa

# Research Aim & Objectives

**To investigate the impacts of climate change on landscape connectivity for the African Savanna elephants and giraffes in KALARIVA TFCL along Balinsky's (1962) African drought corridor between East and Southern Africa under RCP 8.5 scenario.**

1. To identify currently climatically suitable habitats for both African savanna elephants and giraffes in the study area, and in 2100 under RCP 8.5 scenario.

**2. To examine the influence of current land use and land cover on landscape connectivity and the spatial distribution of existing wildlife corridors and pinch points for the focal species in the study area.**

**3. To assess the status of landscape connectivity for the African savanna elephants and giraffes in the study area.**

**4. To map and assess the optimal location of the 'drought corridor' (KALARIVA) in the study area under future climate change scenario in 2100 using under RCP 8.5.**

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# Climatically Suitable Habitats for Giraffes in east and southern Africa-KALARIVA (current and future - 2100) under RCP 8.5

## Research questions

- 1.1 What factors define a suitable habitat, and to what extent does each factor influence habitat selection for giraffes in the study area?
- 1.2 Which areas are currently suitable for the giraffes, and how would habitat suitability patterns for the focal species differ by 2100 under RCP 8.5 scenario?

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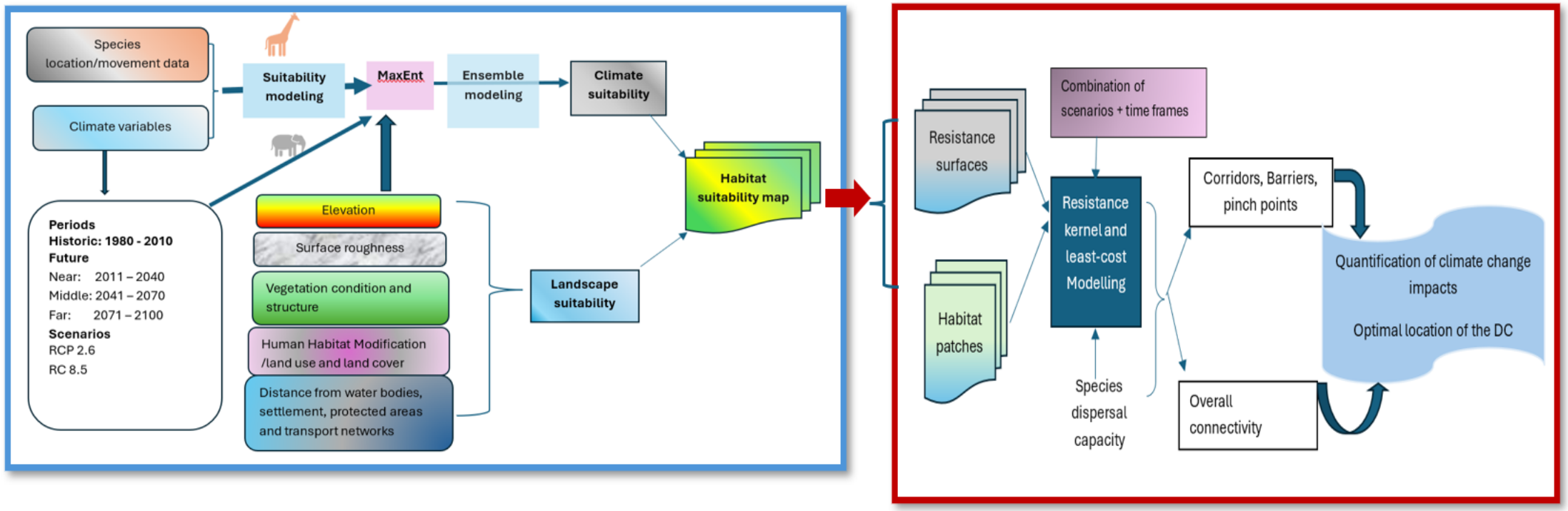
# Propose Methodology

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# Habitat Suitability - Resistance surface layer

- Currently working on a manuscript from Paper 1: Climatically suitable habitats for the giraffes in the KALARIVA: Preliminary results available – WorldClim Dataset
- Data collection: datasets required for the MaxtEnt model ???(Table below)

| Dataset | variable | Status |
|---|---|---|
| Climate | Mean monthly temperature and precipitation | Ongoing |
| Telemetry (GPS) | Species location/movement (2017 – 2024) | Data for giraffes acquired, cleaned, and mapped |
| Topography | Elevation (DEM) | Done |
| Vegetation | Condition (NDVI and EVI) – Woody vegetation cover | Done |
| Global Human Footprint | Human Footprint Index (Global scale) | Done |
| Global Modification Index | Land –uses: Waterbodies, transport networks, settlements (Global scale) | Done |
| Land use land cover | Spatial extent of land uses and land covers | Done |

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# Data and Challenge

- Mean Annual Temperature
- Mean Annual precipitation

Baseline: 1981 – 2010

Near Future: 2011 – 2040

Middle Future: 2041 – 2070

Far/End Future: 2071 – 2100

- CMIP 5
- 3 GCMs and 3 RCM
- RCP 8.5

- Downscale CMIP5 data to 1 Km ---Worldclim dataset????
- MaxEnt- ???

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